TopoLight 0.4

Technical datasheet

Self-hosted network monitoring with a live LLDP/CDP topology map · one static binary · offline licence

Edition covered: 0.4.x (Free, Pro, Team share one binary). Numbers marked "measured" come from the 1,500-device simulated estate used for release testing; nothing else is estimated.

Self-hosted network monitoring with a live LLDP/CDP topology map. One static binary, embedded storage, offline licence. This sheet lists exactly what TopoLight collects, what it shows, what it needs, and what it does not do yet — so you can compare it with other tools on facts.

1. Data sources and protocols

Source	What TopoLight collects	Notes and limits
ICMP echo	Round-trip time, packet loss, jitter per device, every poll cycle (default 60 s); ping-only devices (no SNMP) for servers, printers and appliances	Raw sockets on Linux (`cap_net_raw` set by the installer, or the unprivileged ping sysctl). macOS/Windows builds run SNMP-only reachability.
SNMP v2c	Read-only community polling and discovery	Community stored on disk (mode 0600), never returned by the API.
SNMP v3	authPriv / authNoPriv / noAuthNoPriv; auth SHA-1, SHA-256, MD5; priv AES-128, DES	Own BER codec and USM implementation, tested against net-snmp.
SNMPv2-MIB / HOST-RESOURCES	sysDescr, sysObjectID, sysUpTime (reboot detection), sysName, sysLocation; hrProcessorLoad and hrStorage as CPU/memory fallback	Reboot = uptime went backwards *and* new uptime < 24 h (no false alarms on counter wrap).
IF-MIB / ifXTable	Per interface: name, alias, type, speed, MAC, admin/oper status; 64-bit octet counters → bit/s and utilisation; unicast packet counters → packets/s; ifInErrors/ifOutErrors → errors/s; ifInDiscards/ifOutDiscards → drops/s	32-bit counters used when the agent has no ifXTable, with wrap handling. Interface kinds (physical, LAG, VLAN, tunnel, loopback) classified from ifType.
ENTITY-MIB	Chassis serial number and model	Used for inventory and for LLDP chassis matching.
Vendor MIBs (built-in profiles)	CPU, memory, temperature, sessions: Cisco IOS/IOS-XE, NX-OS, ASA/Firepower; Fortinet FortiGate; Palo Alto; Juniper; Aruba AOS-S and AOS-CX; MikroTik; Huawei; Ubiquiti; net-snmp hosts	Profiles are JSON; drop your own in `<data>/profiles/` to add a vendor without touching the poller.
LLDP-MIB	Local port table, remote chassis ID/port ID/system name/description	Every observation is kept with a confidence score; a link seen from both ends scores 1.0, one end 0.8, an unknown neighbour 0.6.
Cisco CDP (CISCO-CDP-MIB)	Neighbour device ID, port, platform, address	Enabled by the Cisco profiles; feeds the same link builder as LLDP.
Syslog UDP + TCP (514), TLS (6514)	RFC 3164 and RFC 5424 messages; RFC 6587/5425 octet-counting and newline framing; TLS with the console or a self-signed certificate, optional client-certificate requirement	Lines are matched to devices by source address; unknown senders are counted so you can find devices that talk but were never discovered.
SNMP traps and informs (v2c + v3, port 162)	linkDown/linkUp, coldStart/warmStart, authenticationFailure, BGP and ENTITY notifications, plus a vendor-agnostic fallback that keeps raw varbinds; v3 authenticated and decrypted with the saved credentials (keys localised per sender engine ID)	Informs are acknowledged (v3 ones with an authenticated response).

Source	What TopoLight collects	Notes and limits
NetFlow v5 / v9, IPFIX (UDP 2055)	Flow records → per-exporter per-minute and 5-minute summaries: top talkers, targets, conversations, applications (protocol + port), interface in/out	Own decoder; v9/IPFIX templates cached per exporter + observation domain (1 h TTL). Exporters are matched to devices by source IP; unknown exporters still show by address.
sFlow v5 (UDP 6343)	Flow samples with raw Ethernet headers (802.1Q/QinQ aware), scaled by the sampling rate; expanded samples supported	Counter samples are ignored — SNMP already provides interface counters.
BRIDGE-MIB / Q-BRIDGE-MIB	MAC forwarding tables, per VLAN; Cisco IOS through community indexing (`community@vlan`) or the `vlan-N` SNMPv3 context	Every 5 minutes; bridge ports mapped to ifIndex; uplinks (LLDP/CDP, topology links, other devices' chassis MACs) excluded from placement.
ipNetToMediaTable / ipNetToPhysicalTable	ARP and IPv6 neighbour tables from routers, firewalls and L3 switches	Joined with the forwarding tables to give MAC ↔ IP ↔ switch port; vendor from the embedded IEEE OUI registry (53k prefixes, MA-L/M/S).
BGP4-MIB / CISCO-BGP4-MIB / OSPF-MIB	BGP peers (state, remote AS, uptime, last notification decoded, accepted prefixes), OSPF neighbours and router ID, route count	Every 5 minutes; events for session and adjacency changes and for prefix drops.
Q-BRIDGE (VLANs) / BRIDGE-MIB (STP) / IEEE8023-LAG-MIB	VLANs with member ports, root bridge, root port and cost, forwarding/blocking ports, topology-change counter, LAG bundles and member state	Root changes and member loss raise alerts.
UniFi Network API / Cisco Meraki Dashboard API	Access points, switches, gateways with clients per radio and SSID, channel, width, tx power, utilisation, firmware, satisfaction; UniFi WAN1/WAN2 state; Meraki MX uplinks with loss and latency	Read-only user / API key; imported devices are managed by the controller (not polled) and count towards the cap.
AIRESPACE-WIRELESS-MIB / WLSX-WLAN-MIB	Cisco WLC (AireOS, Catalyst 9800) and Aruba mobility controller AP tables: name, IP, model, serial, status, radios with channel, power, clients, utilisation; clients per SSID	APs become managed devices under the controller; no extra credentials.
FORTINET-FORTIGATE-MIB SD-WAN	Health-check members: state, latency, jitter, packet loss per interface	`sdwan_link_down` / `sdwan_degraded` alerts; latency/jitter/loss history.
gNMI (OpenConfig) — beta	`/system/state`, `/system/cpus`, `/system/memory/state`, `/interfaces` over gRPC (TLS or h2c), JSON_IETF: hostname, uptime, version, CPU, memory, interface status, speed, MAC, 64-bit counters	Standard-library HTTP/2 + hand-coded protobuf (no gRPC runtime); Get only, no streaming; tested against grpcio.
SSH (configuration backup)	Running configuration pulled from Cisco IOS/IOS-XE, NX-OS, ASA, FortiGate, PAN-OS, Junos, Aruba AOS-S/CX, MikroTik, Huawei, EdgeOS and Linux hosts on a schedule and after "config changed" syslog lines; versions with normalised comparison, side-by-side line diff, download; `config_backup_changed` / `config_backup_failed` events	Password or private key, optional enable; read commands only; gzipped history, 50 versions per device.
Synthetic probes	TCP connect, HTTP(S) status/body, DNS answer, TLS handshake and certificate expiry, ICMP ping, traceroute	Run from the TopoLight host on a schedule; latency series, `probe_failed` / `tls_expiring` alerts, path-change events; tied to devices optionally.
Reports	Availability/SLA, alerts and MTTR, utilisation and load, inventory, configuration changes, flow, endpoints, probes —	Built from the stored history; scheduled copies kept on disk.

Source	What TopoLight collects	Notes and limits
	HTML (print to PDF) and CSV; daily/weekly/monthly by e-mail	
API (bearer tokens)	Every console endpoint, JSON, with per-token role	Tokens hashed at rest; created and revoked in Admin → Users.
Outbound	SMTP with STARTTLS; Telegram Bot API; webhook with `X-TopoLight-Signature: sha256=<HMAC>`	Grouping window, quiet hours, minimum severity, "critical always" punch-through.

2. What it does with the data

Discovery. Sweeps subnets and ranges (largest /20 per line) with ICMP then SNMP using every saved credential; follows LLDP/CDP neighbours automatically; manual add for anything else. Devices beyond the licence cap are listed as *not monitored*, never silently dropped.

Topology. Links are synthesised from both ends of every LLDP/CDP observation. Roles (core, distribution, access, router, firewall) are inferred from the graph — a redundant core pair counts as two — and can be locked by hand. The map is a 3D stacked-disc view drawn on a plain canvas (no WebGL, no JavaScript framework) with a 2D mode, orbit, zoom, status and utilisation overlays, and live updates over a server-sent event stream. Links that disappear are kept greyed for 7 days, then dropped; manual links are never dropped.

Health at a glance. Hovering a node shows a health card: status and cause, CPU, memory, temperature, RTT and loss, interface counts (up / down / shut, uplinks down), aggregate traffic in and out as bit/s and packets/s, drops and errors per second, the three busiest ports, and the ports that are admin-up but operationally down. The same card is in the click-through panel and the device page.

Flow. Who talks to whom: a Flow page and a per-device Flow tab with a throughput chart and top-N tables for talkers, targets, applications, exporter interfaces (named from the SNMP inventory) and conversations, over 5 minutes to 24 hours. Bounded by design — top 100 / 200 / 50 per bucket, ≈2 MB per exporter per day of history once gzipped (≈26 KB per 5-minute summary while the day is open) — so it never becomes a storage problem. Config snippets include the export commands for each vendor.

Endpoints. Where is 10.10.20.15 plugged in? Every MAC seen on the network, with its IPs, vendor, switch, access port, VLAN, first/last seen and port moves — searchable by any of those, grouped per port on the device page, with `endpoint_moved` events. Bounded to 200k MACs / 90 days.

State engine. Per-device and per-interface state machines with confirmation cycles (default: down after 3 failed cycles, up after 2 good ones), separate enter/exit thresholds so a value hovering at the line does not flap, flap detection (more than 5 changes in 10 minutes), and **topology-aware suppression**: when a device goes down, everything whose only path runs through it becomes *unreachable* and its alert folds under the root cause. More than 80 % of a site down collapses into one site alert. Maintenance windows silence devices without deleting anything.

Alert rules (all editable). device down, SNMP unreachable, CPU, memory, temperature, ICMP loss and latency, interface utilisation and error rate, link down (trap/syslog first, confirmed by a poll), reboot, configuration change, BGP/OSPF neighbour, HA state change, environment fault, authentication failures, log flood, VPN tunnel change. Resolved alerts that re-trigger within 30 minutes re-open with their history instead of duplicating.

Console. Overview, topology, alerts with keyboard navigation (`j`/`k`/`a`/`r`), device pages with 24-hour graphs, log explorer with histogram, admin for sites, credentials, notifications, rules, maintenance, users and licence. Dark and light themes, WCAG AA (0 axe violations on 18 page/theme combinations at release), command palette on `/`.

Security. Login required from the first request; PBKDF2-SHA256 (600k) passwords; HttpOnly + SameSite=Strict sessions; same-origin enforcement on state-changing requests; CSP forbidding inline and remote scripts; credentials never leave the data directory. TopoLight is read-only towards devices — it never pushes configuration.

3. Editions

	Free	Pro	Team
Monitored devices	25	500	1,500
Sites	1	3	unlimited
Metric and log retention	7 days	6 months	12 months
Users	3	5	unlimited
Everything in §1–2, plus Telegram and signed webhook, maintenance windows, admin / operator / viewer roles, JSON export API	✓	✓	✓
Price	$0 (GitHub)	$49 / month	$149 / month

One binary for all three and no feature gating — tiers differ only in capacity and history; an offline Ed25519 licence key raises the caps. Over the cap, the newest devices are marked *not monitored* and the API answers `402` with a readable message — nothing breaks, nothing is deleted. A missing or expired key runs as Free.

4. Server requirements

TopoLight is **one process on one host**. The binary contains discovery, the ICMP pinger, the SNMP poller, the syslog and trap receivers, the topology builder, the state and alert engine, the notifier, the embedded time-series and log stores, and the web console. Nothing is installed on the monitored devices; no database server, message bus, cache or agent is installed next to it.

Measured on a 1,500-device simulated estate polled every 60 s: about 0.25 CPU cores on average and ~110 MB RSS. Disk for history is the real cost: **≈4.5 KB per series per day at 60-second resolution** and **≈1.8 KB per series per day at 5-minute resolution** (worst case, noisy values). Raw 60-second samples are kept 7 days, then 5-minute avg/min/max to the retention limit; non-uplink ports are stored at 5-minute resolution from the start.

Estate	Host	History on disk (worst case)
100 devices, 24-port switches, 6 months	1 vCPU · 1 GB RAM	~2 GB
500 devices, 48-port switches (Pro, 6 months)	2 vCPU · 2 GB RAM	~17 GB
1,500 devices, 48-port switches (Team, 12 months)	2 vCPU · 4 GB RAM	~100 GB

Idle access ports compress far better than the worst case; real estates typically land at a third of those figures. Admin → System shows the live number.

Network. Outbound UDP/161 to devices; inbound UDP/514 + TCP/514 (syslog), UDP/162 (traps), UDP/2055 (NetFlow/IPFIX) and UDP/6343 (sFlow) from devices; TCP/8433 for the console (put TLS or a reverse proxy in front, or use `-tls-cert`/`-tls-key`). Linux amd64/arm64 for production; macOS and Windows builds for trying it on a laptop.

Deployment modes

Mode	Status	What it means
Standalone	shipped	One host monitors the whole estate over routed IP.
Cluster — standby nodes	shipped	2–5 servers joined with one command; every full node holds a complete mirrored copy (10-second refresh); one leader runs

Mode	Status	What it means
		the engine, the others poll their share and forward. Three or more full nodes: automatic leader election in ~20 s. Two: manual promote. Any node's console works.
Cluster — collectors	shipped	Poll-and-forward nodes for branches behind NAT or slow WAN: pin a site to a collector; it polls locally and forwards samples, syslog, traps and flows.
Warm standby (host-level)	works, manual	Hypervisor HA or a copy of `/var/lib/topolight`; still the simplest answer for a single small site.

Honest limits: one leader runs the one state engine, so a cluster raises polling throughput and survives a dead server — it does not turn TopoLight into a 50,000-device platform. Failover loses at most ~10 s of logs and ~1 minute of metrics. Nodes need TCP/8434 to each other.

Sizing per node

Scenario	Node	Notes
Standalone ≤ 500 devices	2 vCPU · 2 GB · 40 GB SSD	measured on 0.1 (¼ core, 110 MB RSS at 1,500 devices)
Standalone ≤ 2,000 devices + flow	4 vCPU · 4 GB · 250 GB SSD	flow summaries ≈2 MB/exporter/day; configs and endpoints are small
Cluster 3 full nodes ≤ 2,000 devices	3 × (2 vCPU · 4 GB · 250 GB)	full copy on each node; automatic failover
Branch collector	1 vCPU · 1 GB · 10 GB	holds only the inventory snapshot

5. How it compares

Stated as differences, not verdicts — each of these tools is good at things TopoLight does not attempt.

	TopoLight 0.4	LibreNMS	Zabbix	PRTG
Install footprint	1 static binary, embedded stores, one directory to back up	PHP + MySQL/MariaDB + RRD + poller workers	Server + database (PostgreSQL/MySQL) + frontend + agents/proxies	Windows core server + probes
Topology	Drawn from LLDP/CDP with confidence scores; roles inferred; 3D/2D live map	Discovery-based maps and plugins	Manual/network maps, discovery rules	Auto-discovery, sensor tree, maps
Root-cause handling	Graph-based suppression built into the state engine (unreachable ≠ down)	Dependency-based parent/child	Trigger dependencies, configured per trigger	Sensor dependencies
Traffic data	SNMP/gNMI counters (bps, pps, drops, errors), NetFlow/IPFIX/sFlow top-N	SNMP + NetFlow/sFlow via plugins	SNMP + collectors	SNMP, flow, packet sniffer sensors
Pricing model	Per instance: 25 devices free, $49 (500), $149 (1,500); offline	Free (GPL)	Free (AGPL) + paid support	Per sensor

	TopoLight 0.4	LibreNMS	Zabbix	PRTG
	key			
Breadth	Network devices only, on purpose	Very broad	Very broad (servers, apps, cloud)	Very broad

6. Not yet — public roadmap

gNMI streaming subscriptions (Get only today) · BMP · Docker Hub image · more controller APIs (Aruba Central, Mist, Omada). Against a full enterprise NMS/observability catalogue (availability, health, flow, packet, L2 topology, routing, streaming telemetry, logging, identity, network services, wireless/SD-WAN, OAM, cloud) TopoLight covers availability (incl. synthetic checks), topology, device/interface performance, endpoints, flow, routing state, wireless/SD-WAN health and configuration backup; event logging and streaming telemetry partially; and none of the rest — that is the scope of a tool built for 25–2,000 network devices, not a platform claim.

7. Links

GitHub (free edition, source, issues): github.com/nizartuanku/topolight · Whop (Pro/Team, 14-day trial): whop.com/nizar-tuanku/topolight · Docs: `docs/INSTALL.md`, `docs/USER-GUIDE.md`, `CHANGELOG.md`. Part of the Hexward line of self-hosted tools.